Networking

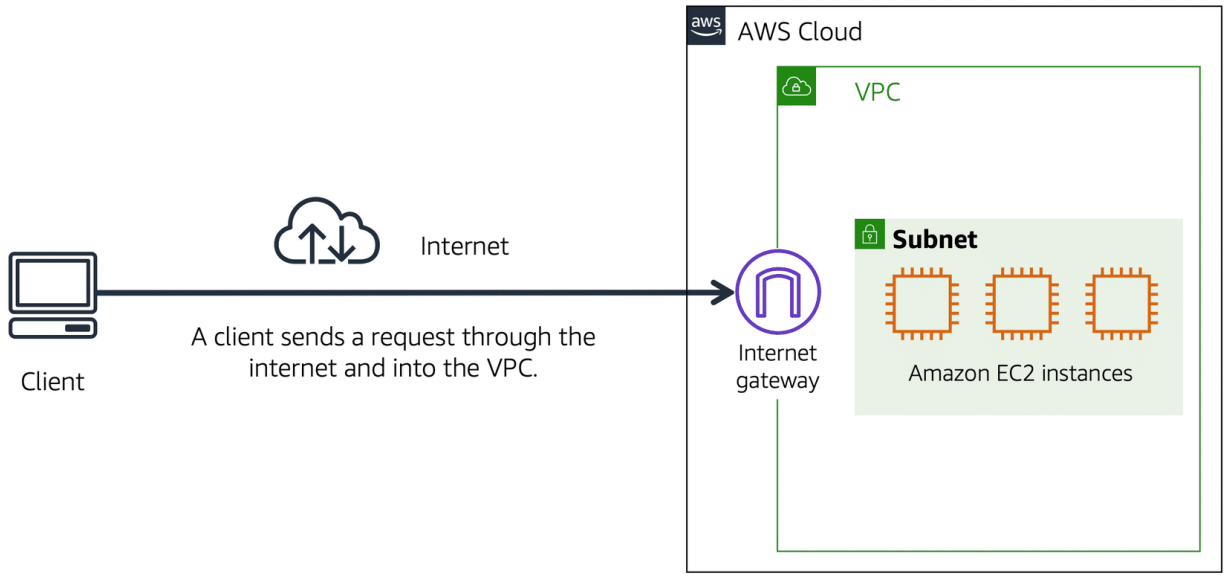
Amazon Virtual Private Cloud (VPC)

- A networking service that you can use to establish boundaries around your AWS resources
- Amazon VPC enables you to provision an isolated section of the AWS Cloud.
- In this isolated section, you can launch resources in a virtual network that you define.
- Within a virtual private cloud (VPC), you can organize your resources into subnets.

Internet Gateway

To allow public traffic from the internet to access your VPC, you attach an **internet gateway** to the VPC.

An internet gateway is a connection between a VPC and the internet.

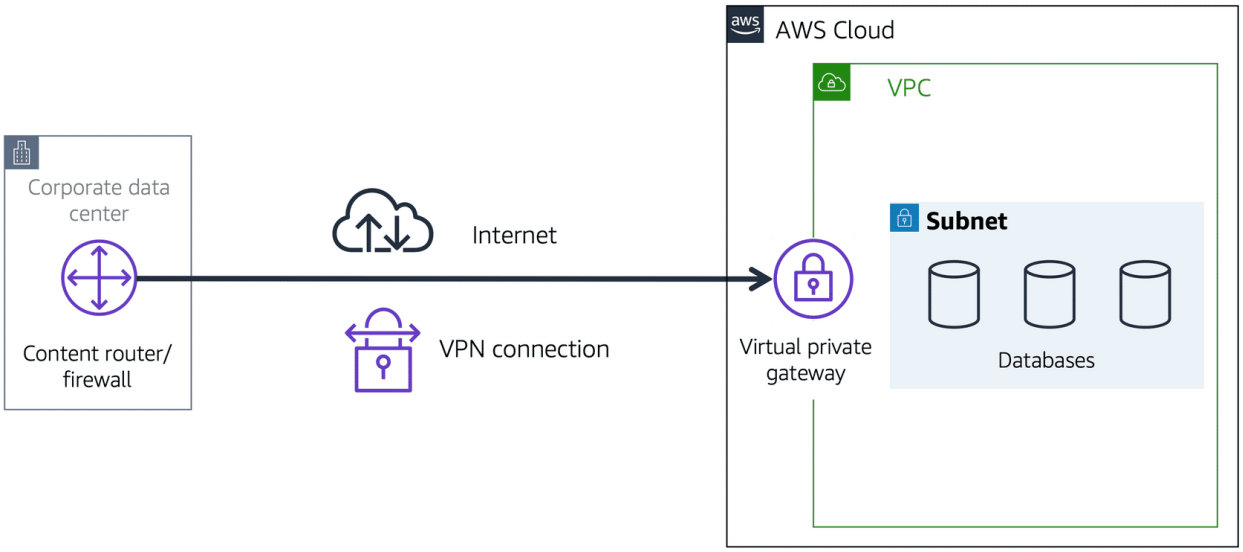


What if you have a VPC that includes only private resources?

Virtual private gateway

To access private resources in a VPC, you can use a **virtual private gateway**.

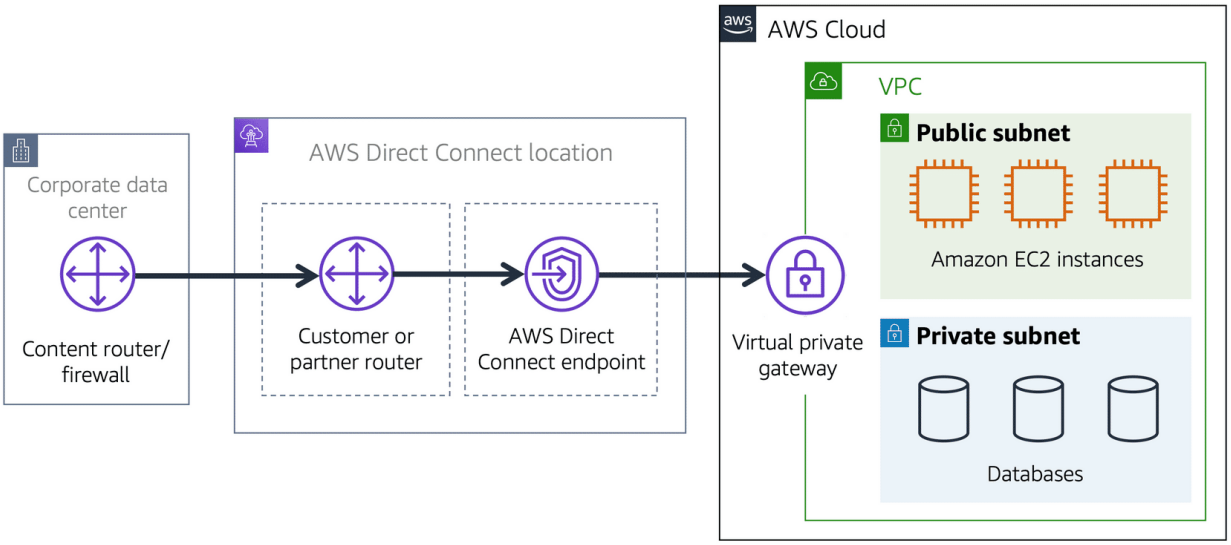
A virtual private gateway enables you to establish a virtual private network (VPN) connection between your VPC and a private network, such as an on-premises data center or internal corporate network. A virtual private gateway allows traffic into the VPC only if it is coming from an approved network.



AWS Direct Connect

service that enables you to establish a dedicated private connection between your data center and a VPC.

The private connection that AWS Direct Connect provides helps you to reduce network costs and increase the amount of bandwidth that can travel through your network.



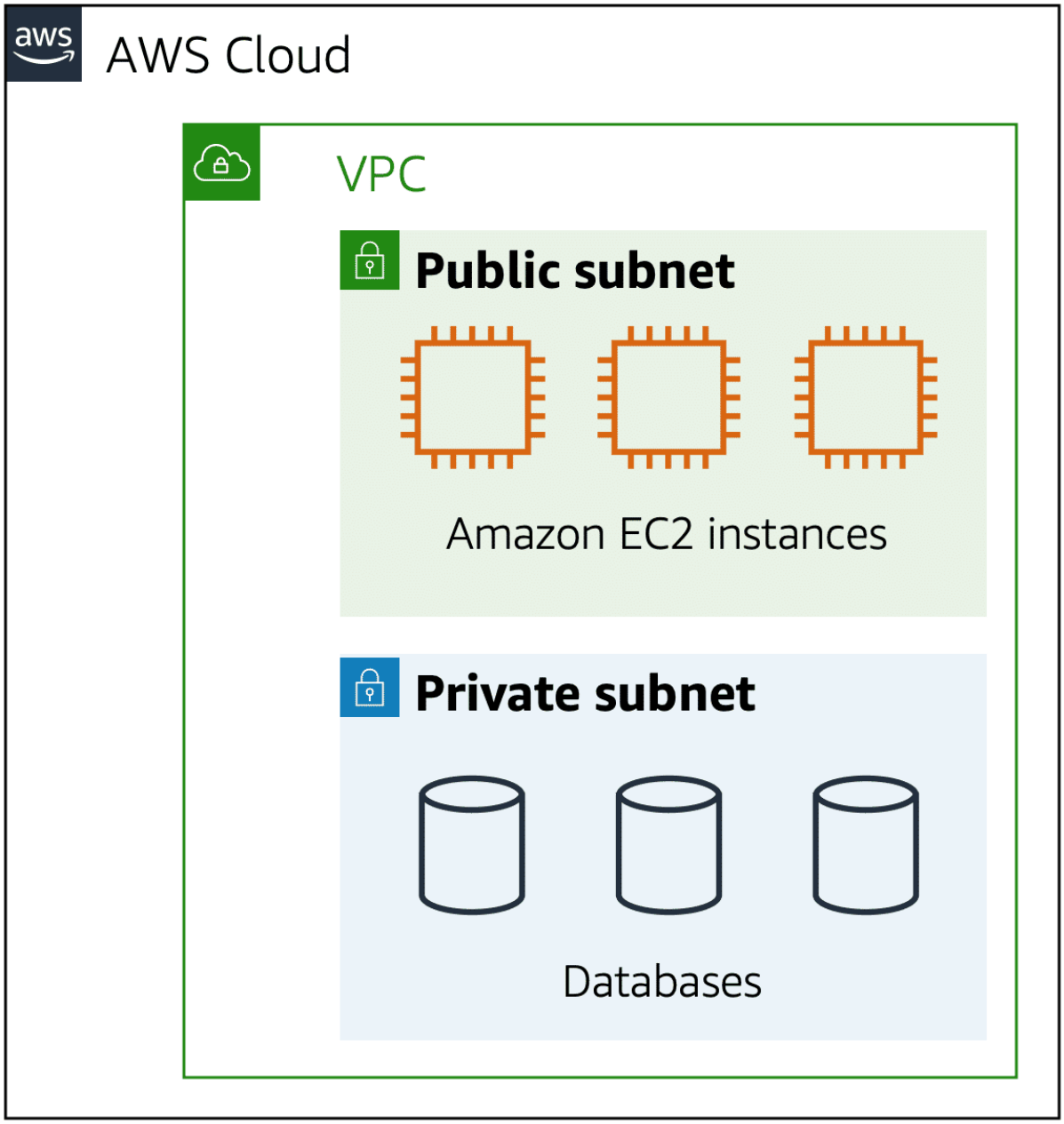
Subnets

A subnet is a section of a VPC in which you can group resources based on security or operational needs. Subnets can be public or private.

Public subnets contain resources that need to be accessible by the public, such as an online store's website.

Private subnets contain resources that should be accessible only through your private network, such as a database that contains customers' personal information and order histories.

> In a VPC, subnets can communicate with each other. For example, you might have an application that involves Amazon EC2 instances in a public subnet communicating with databases that are located in a private subnet.



Network traffic in VPC

When a customer requests data from an application hosted in the AWS Cloud, this request is sent as a packet.

It enters into a VPC through an internet gateway. Before a packet can enter into a subnet or exit from a subnet, it checks for permissions. These permissions indicate who sent the packet and how the packet is trying to communicate with the resources in a subnet.

The VPC component that checks packet permissions for subnets is a Network Access Control List (NALs)

Network Access control List (NALs)

virtual firewall that controls inbound and outbound traffic at the subnet level.

Each AWS account includes a default network ACL. When configuring your VPC, you can use your account's default network ACL or create custom network ACLs.

By default, your account's default network ACL allows all inbound and outbound traffic, but you can modify it by adding your own rules.

For custom network ACLs, all inbound and outbound traffic is denied until you add rules to specify which traffic to allow.

Additionally, all network ACLs have an explicit deny rule. This rule ensures that if a packet doesn't match any of the other rules on the list, the packet is denied.

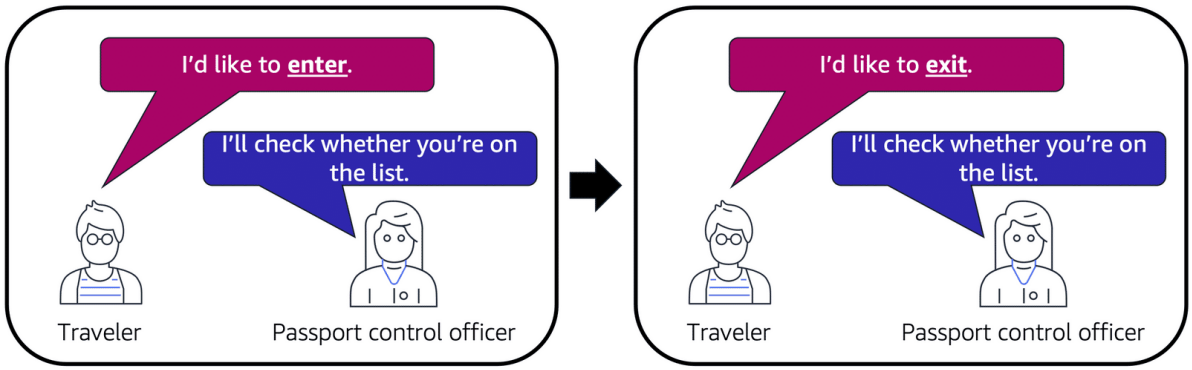
Stateless packet filtering

Network ACLs perform **stateless** packet filtering. They remember nothing and check packets that cross the subnet border each way: inbound and outbound.

When a packet response for that request comes back to the subnet, the network ACL does not remember your previous request. The network ACL checks the packet response against its list of rules to determine whether to allow or deny.

After a packet has entered a subnet, it must have its permissions evaluated for resources within the subnet, such as Amazon EC2 instances.

The VPC component that checks packet permissions for an Amazon EC2 instance is a Security Groups



Security groups

A security group is a virtual firewall that controls inbound and outbound traffic for an Amazon EC2 instance.

By default, a security group denies all inbound traffic and allows all outbound traffic.

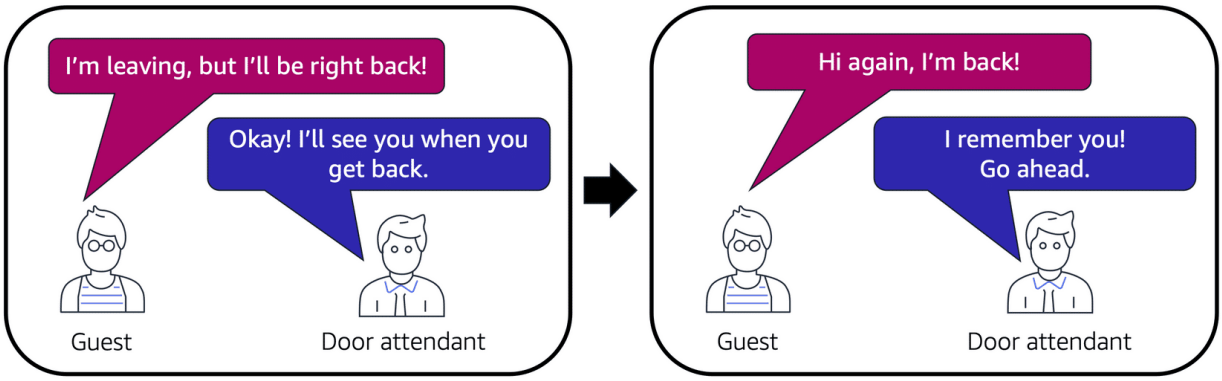
You can add custom rules to configure which traffic to allow or deny.

If you have multiple Amazon EC2 instances within a subnet, you can associate them with the same security group or use different security groups for each instance.

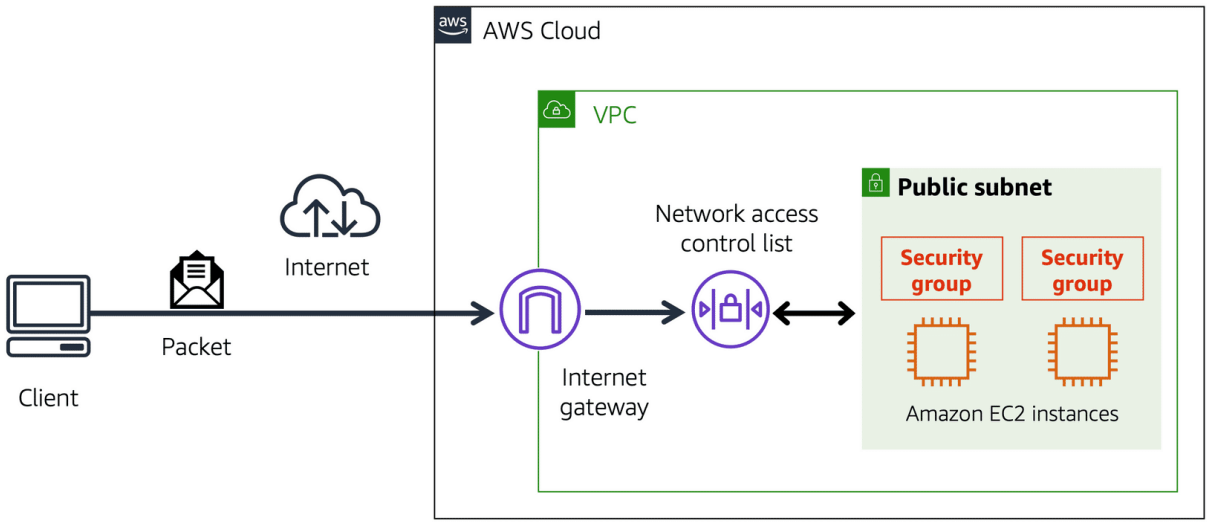
Stateful packet filtering

Security groups perform **stateful** packet filtering. They remember previous decisions made for incoming packets.

When a packet response for that request returns to the instance, the security group remembers your previous request. The security group allows the response to proceed, regardless of inbound security group rules.



Both network ACLs and security groups enable you to configure custom rules for the traffic in your VPC. As you continue to learn more about AWS security and networking, make sure to understand the differences between network ACLs and security groups.

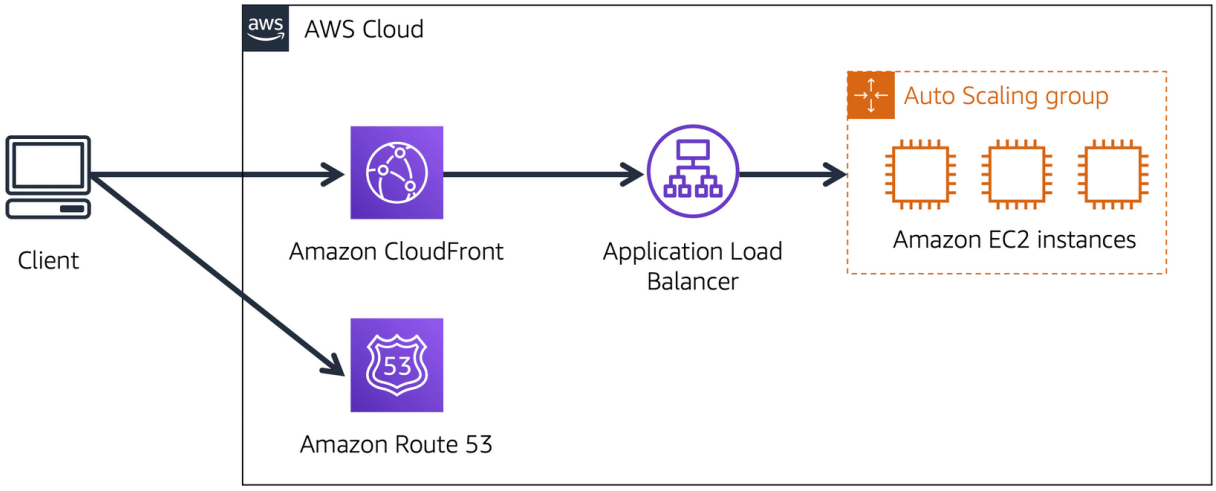


Amazon Route 53

is a DNS web service. It gives developers and businesses a reliable way to route end users to internet applications hosted in AWS.

Amazon Route 53 connects user requests to infrastructure running in AWS (such as Amazon EC2 instances and load balancers). It can route users to infrastructure outside of AWS.

Another feature of Route 53 is the ability to manage the DNS records for domain names. You can register new domain names directly in Route 53. You can also transfer DNS records for existing domain names managed by other domain registrars. This enables you to manage all of your domain names within a single location.



1. A customer requests data from the application by going to AnyCompany's website.
2. Amazon Route 53 uses DNS resolution to identify AnyCompany.com's corresponding IP address, 192.0.2.0. This information is sent back to the customer.
3. The customer's request is sent to the nearest edge location through Amazon CloudFront.
4. Amazon CloudFront connects to the Application Load Balancer, which sends the incoming packet to an Amazon EC2 instance.